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The Machine of the Mind

Part I. The computer and the brain.

Author: Renato Nobi
Department of Physics "G. Galilei"
University of Padova - Italy

Summary

In this paper I will deal with some topics concerning artificial intelligence. After a brief historical analysis of the Boolean logical paradigm, I will introduce some theoretical arguments which are essential for understanding how the self-reflexive capacity of human thought can be produced and I will try to explain the reasons that lead to consider the human brain as a *universal recursive machine*. Finally, I will compare the working principles of serial calculation with those of parallel calculation to try to understand if there are substantial differences of principle between the processing capacity of computers and those of animal brains.

In a second article (entitled *The Machine of the Mind, Part II – The role of synchronization in brain information processing*), I will carry out a survey on the main anatomical and functional aspects of the mammalian brain, in order to describe the most interesting discoveries that have taken place during the past two decades of the last century and the perspectives that have opened up in understanding the processes that underlie the functioning of cognitive memory.

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1.1. The dream of Boole

For about forty years after the publication of the article of McCulloch and Pitts *A Logical Calculus of Ideas Immanent in Nervous Activity* (1943), the neuron was modeled as a *threshold logic unit*, i.e., an element with n inputs and one output, that generates a step-shaped output when an algebraic sum of positive or negative input signals exceeds a certain value, called the threshold of the neuron (Fig.1). Throughout that period the complexity of the nervous processes was underestimated and the potentialities of electronic calculus overestimated.

For these reasons, the mode of operating of the neuron could appear as elementary as that of a thermionic valve or a transistor, the discovery of which, let us remember it, dates back to the early 1950s. Since at that time it was ignored that the supporting tissue of the nerve network (the glia) exchanges signals with the neurons – performing for these important regulatory functions –

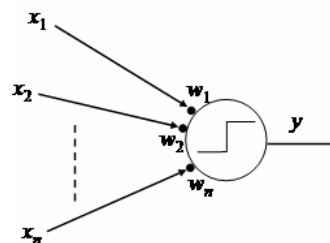


Fig.1. The neuron as a threshold logic unit:

*If $x_1 + x_2 + \dots + x_n > \text{threshold}$, $y=1$,
otherwise $y = 0$.*

$w_1, w_2, \dots, w_n$ = transmission coefficients (weights) of the synapses.

the exchange of signals seemed to be carried out by nerve cells only. It was therefore natural to think that the mental processes should be ascribed to the interaction of functional units similar to the elementary components of an electronic circuit (Arbib, 1987). The analogy seemed all the more correct in that the nerve signals were believed to propagate electrically through synaptic contacts regarded as mere welding points of nerve cells. The location of the memory, on the other hand, appeared problematic. Some believed that nervous information is stored in the threshold values of neurons, others that it is encoded in molecules contained in neurons, perhaps in the DNA itself. Of the various hypotheses that were formulated in those years, the one that survives today is that of Hebb (1949), who hypothesized that memory is due to the formation or alteration of the synaptic contacts through which the cells of a neuronal population communicate. To understand what was the difference between this view and that of today, consider that now the functional complexity of a single neuron is estimated to be comparable to that of a microprocessor. Actually, this comparison is partly exaggerated and partly wrong, because the single neurons do not possess significant computational capabilities and are variously specialized for carrying out different functions.

The most appealing aspect of the optimistic view of McCulloch and Pitts was the systematic use of a model of neuron with *all-or-nothing* response. This simple device had its theoretical legitimacy in the representation of numbers in binary digits (*bits*), i.e., as ordered arrays of **0** and **1** regarded as sequences of dichotomic decisions (Shannon, 1949). This assumption corresponded perfectly to what might have seemed the prophetic legacy of George Boole (1853), the inventor of the *logical calculus*, who introduced the idea that the basic rules of the logical language [*conjunction* (AND), *disjunction* (OR), *negation* (NOT) and their combinations] – or rather the fundamental operations carried out by the brain – formed a sort of algebra of two-valued variables identified as propositions that may take the values: *all* = **1** = *true*; *nothing* = **0** = *false*.

In fact, if we assume that the elementary constituents of the nerve network function as threshold decision units, it is natural to hypothesize that the processes occurring in the brain are similar to those of the logical calculus of Boole.

Let us imagine the fun of McCulloch and Pitts and after them of Kleene (1956), who reconsidered the subject in a more complete and rigorous way by proving that all the logical operations of the Boolean algebra can be carried out by suitable combinations of time-delayed threshold logic units, as shown in Fig.2.

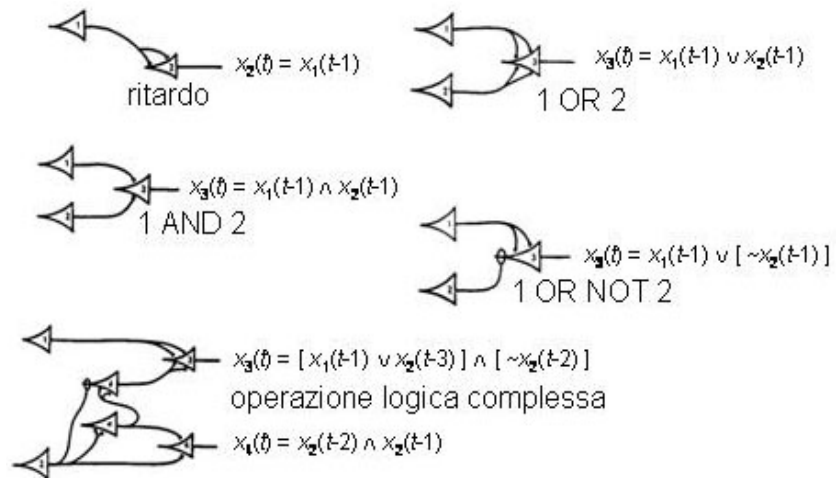


Fig.2. Some neuronal circuits described by McCulloch and Pitts, their Boolean operational equivalents and the corresponding logical formulas.

Moreover, since the operation of a threshold logic unit can be approximated as well as desired by a circuit built up with elementary Boolean operators and time delay devices, it was natural to infer that any information process carried out by combinations of threshold logic unit could be simulated by a sufficiently complicated network of Boolean operators. Therefore, if the neural networks of McCulloch and Pitts are good models of real nervous networks, then the animal brains and the computers would be two different but functionally equivalent ways of processing information; with all the consequences that we can imagine for both the technological and the philosophical levels. The impetuous development of information technology of the 1960s contributed to enhance this illusion of simplicity (De Luca and Ricciardi, 1981).



Fig.3. Kurt Gödel
(1906-1978)

In reality, as proven by Alan Turing in 1936, in order to achieve large computing capabilities, a network of logical operators alone is not sufficient, but they are also necessary devices capable of writing, reading, cancelling and carrying out arithmetic operations, so as to make it possible to execute *recursive procedures*, i.e., ordered sets of logical and mathematical operations instructed by programs of finite length. By recursive procedures applied to finite or infinite sets of data it is possible to implement mathematical calculations of any sort and degree of complication, so as to achieve the *computational universality*; that is, the capability of simulating any computer.

In this way, the question of the computational capability of recursive processes could naturally connect to the interesting problem of the relation between *logic* and *arithmetic*: in particular, to the theorems of decidability and undecidability of mathematical propositions – discovered by Kurt Gödel (1930, 1931) – and also to the theory of the relationship between *language* and *metalanguage* of Alfred Tarski (1936). Thus, it seemed that the "quadrature" of the brain could also be achieved: all that a human brain can do, can be simulated by a sufficiently powerful computer.

If anything, the problem was to explain how the brain could perform very precise calculations by means of unreliable and perishable devices as neurons are. In this regard von Neumann proved that this difficulty can be overcome by increasing the redundancy of the neuronal circuits.

Basing on the ideas of Turing, von Neumann planned and built the ENIAC, the first universal electronic computer. Thus, the Boolean investigation on the laws of thought seemed to end in a perfect circle: *Thought is logical because the functional elements of the brain are similar to those of the computers.*

Could anyone desire more? What doubts could have arisen about the accuracy of this approach, considering that mathematicians, of the level of Shannon, Moore, Arbib and others, contributed to legitimate and elevate this view to the rank of a complete systematic theory?

These ideas guided for many years the research on *artificial intelligence* (AI), about which it is worth mentioning the names of Minsky, Newell, Simon. Unfortunately, the results in this area have been disappointing. In forty years of intense research the software produced by AI scholars has never provided something that may vaguely resembles an implementation of human thought. What were the causes of these difficulties? In this paper I will try to answer this question.

1.2. Gödel, Turing e von Neumann

For a few decades, some logicians and theorists of AI fed an ambitious project: explaining how an information processing can be capable of "self-reflection", i.e., capable of reproducing our capability of thinking what we have thought; in others words, of generating our self-awareness.

This topic was disclosed in a compelling and suggestive way by Douglas Hofstadter in the book *Gödel, Escher, Bach* (1979). This author was inspired by the self-referential properties of arithmetic discovered by Gödel, but also,

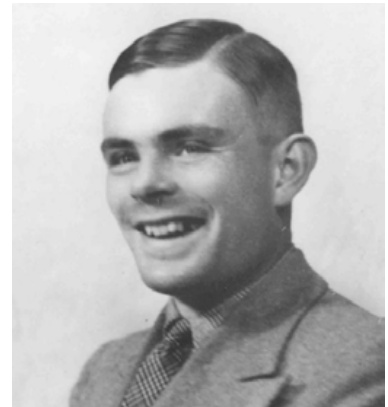


Fig.4. Alan Mathison Turing
(1912-1954)



Fig. 5. John von Neumann (1901-1956)
and the first computer ENIAC
(*Electronic Numerical Integrator and Computer*).

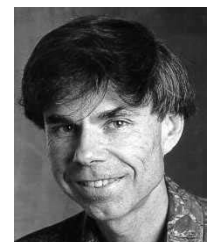


Fig.6. Douglas
Hofstadter (1945 – ...)

(although the author does not declare it explicitly) from the analogous theory of machines equipped with universal construction capabilities, which was outlined by John von Neumann in a series of lectures and notes collected by his collaborators between 1949 and 1956 (year of von Neumann's death) and published in 1966 as the *Theory of Self-Reproducing Automata*.

As in the vision of Gödel the self-reference capability of arithmetic depends critically on the condition that this algorithm be capable of doing all that any other algorithm can do, i.e., that it be a *universal algorithm*; thus, it will be capable of carrying out computations of any sort. Similarly, in the vision of von Neumann, the self-reproduction capability of the automaton depends critically on the condition that this machine be capable of doing all that any other machine can do, i.e., that it be a *universal machine*; in which case it will be capable of constructing machines of any type and degree of complexity. The formal analogy of the two visions is evident.

However, in order that such interpretative or constructing capabilities can be really implemented, the system under consideration (the algorithm or the automaton) must be able of carrying out a number of essential functions that unavoidably require a sufficient degree of structural complexity. If only one of these functions were missing, the system would lose the property of universality and would therefore no longer be capable of self-referencing or self-reproducing. This can be easily understood considering that self-reference and self-reproduction are ultimately nothing other than computational and constructive processes. This important topic deserves a rather broad digression.

Although developed in the past century, the concepts that I am going to outline go far beyond the level of the general scientific culture of the past century, which was mainly dominated by ideas and concepts borrowed from physics. But today the effort of their acquisition is worth doing because they are spreading out exponentially in all directions in the immense territory of the *sciences of complexity*: information theory and informatics, the theory of languages, microbiology and sciences of life. To describe the logical sequence of steps necessary to acquire them I will use the Latin ordinals followed by titles indicative of the topics and I will italicize the most important definitions.

I. Algorithms and processes.

Arithmetic and elementary algebra are very popular *algorithms*. The rules for the construction of geometric figures using a ruler and a compass also constitute an algorithm, although in this case we deal with graphical rather than with arithmetic and algebraic operations.

With maximum generality, we can say that an algorithm is a set of rules that can be applied to a dataset to produce results that are partly canceled and partly stored in the same dataset. A sequence of operations applied to the same dataset constitutes a process. A process is called *recursive* if the sequence proceeds deterministically through calculation cycles instructed by a program, so that at each cycle both the initial data and those produced in the previous cycles can be used.

A recursive process can stop at a certain moment, when the desired result has been obtained, or ideally continue forever in order to produce an increasingly approximate result. In mathematical practice, algorithms are symbolic elaborations that allow to produce mathematical, geometric and physical entities useful for science.

II. Computational universality.

Let us say that an algorithm interprets another if, operating on a set of data and symbols with own methods, generally different from those of the other, is capable of producing results equivalent to those of the other. Briefly, if it can *simulate* the behavior of the other. An algorithm that is not complex enough can interpret, at most, similar or simpler algorithms. For example, using exponentials and logarithms it is possible to interpret multiplications and divisions in terms of additions and subtractions. The old ruler of the surveyors, now fallen into disuse, did just that.

An algorithm capable of interpreting any other algorithm called *universal*. In the same vein, a computer capable of simulating any other computer is also called universal. The discovery that a combination of logical and arithmetic operations can generate a universal algorithm is implicit in Gödel's work and was made explicit by Turing with the demonstration of the existence of universal computers. Turing described how to build a machine capable of simulating the calculation procedures of any other calculating machine, even one of much greater complexity.

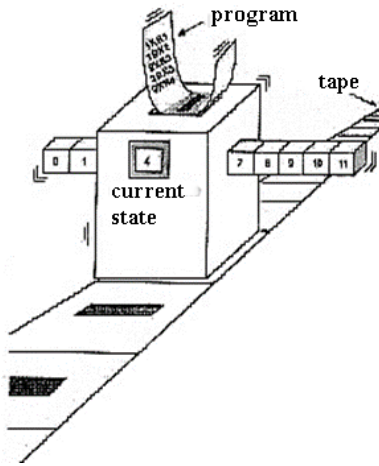


Fig.7. Turing machine.

The Turing machine is equipped with a little head capable of moving on a tape of unlimited length, reading or writing on this tape a number of symbols and storing the data just read, the current position of the head and the internal state of the machine in one or more registers (Fig.7). The head moves back and forth following instructions written in a finite part of the tape, which plays the role of *program*, and writes the results on the remaining part of the tape which therefore functions as *working memory*. The instructions are of the type "if the head has read data *A* and the internal state is *B*, then do this, otherwise do this other".

The general scheme of the functions performed by a Turing machine is illustrated in Fig.8, it also corresponds quite well to the functioning scheme of a common desktop computer

The most relevant aspect of Turing's theory is that *computational universality is equivalent to the capability of carrying out every possible sort of logical and/or possible and imaginable arithmetic calculation*, the price to be paid being only the calculation time. We can therefore clearly understand how important is for mathematics the discovery that *arithmetic is a universal algorithm*. This is the reason why all mathematical procedures – even those that are purely geometric or physical – provided that they are suitably interpreted, can be ultimately reduced to a sequel of logical-arithmetic calculations. Such is, for example, the arithmetic representation of the theorem of Pythagoras is $x^2 + y^2 = z^2$.

The most important result of the theory of Turing is the discovery of the *halting problem*: it states that no universal machine can generally predict whether another universal machine will end its calculation unless by simulating faster the same calculation.

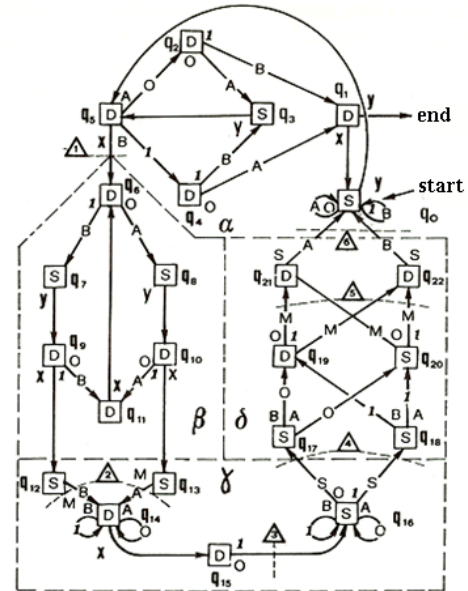


Fig.8. Functional diagram of a universal Turing machine. From Aiello et al. (1976).

III. Logic.

Every proposition of a logical language, whether mathematical or not, has a value of truth. In Boolean logic the proposition can be *true* = 1 or *false* = 0. Combining logical propositions by means of Boolean operations (AND, OR, NOT), we obtain new propositions whose values of truth can be established by using the tables of truth of a basic set of propositions. All the propositions that can be obtained by means of Boolean operations applied to a preestablished set of basic propositions, regardless of their values of truth, are called *well-formed* (in the case of geometry, one may say *well-drawn*).

Each logical theory starts from a certain number of basic propositions, called *axioms*, and aims to prove other propositions, called *theorems*. Axioms are well-formed propositions that are assumed *true* by definition. The purpose of a proof is to evaluate the truth or falsity of a well-formed proposition starting a set of axiomatic truths. The interested theorems are those that are particularly important for the uses that we can be made of them. In the course of a proof, it may happen that certain complicated propositions be replaced by simpler ones or can be simplified by introducing suitable definitions, or using new operations and logical rules.

In practice, *logical operations and rules formalize mathematical reasoning*. They govern a sequence of steps (logical implications) some of which have the effect of increasing the length of the logical proposition (in the case of geometry the complexity of the geometric figure), others of decreasing it, so as to produce a final result – a logical proposition (a geometric figure) – that is considered interesting. Decreases of length can be obtained by applying the *rules of inference of the axiomatic system* – which in practice state that certain mathematical expressions can be replaced by others less complicated – or by applying a fundamental rule of the logic, for example the *modus ponens*, which allows the replacement of expression "*if A then B*" simply with "*B*" (in the case of geometry the cancelation of the part of drawing that was used to produce a certain result). Since the evaluation of a proposition falls outside the logical calculus, it is evident that in reality every logical theory is subordinated to an interpretative system much wider than that of the logic.

However, there is a great difference between the logic of a language that describes a finite world of objects and that of a language capable of describing the properties of mathematics, for example, those of the arithmetic. In this second case, the universe of the discourse contains an infinite number of entities, which often consist of variables of which we want to determine the properties or of unknown quantities of which we want to determine the value. This happens, for example, when we want to find the solution of an equation. In these circumstances, the logical language needs to hypothesize or enunciate the *existence* of an object in an infinite set of possible objects, or to refer to *all the objects* of an infinite set. In this regard it is important to consider that although the data of knowledge and the concrete acts of human action are finite, nevertheless they are inscribed in infinite domains of the *possible*: infinite physical possibilities, infinite possibilities of thought, infinite positions of an object in the space etc.

Statements of existence and references to a totality are called *existential quantifiers*. A logic that does not need existential quantifiers is said to be of the *first order*, one that needs them is called of the *second order*. In practice, a logic of the first order expresses the fact that a statement regarding the variable included in the quantifier is true for at least one value of the variable. It can be put in the form of a collection of propositions whose truths or falsehoods can be orderly ascertained one after the other. The logic of arithmetic is of the second order because it deals with uncountable sets and problems of existence in uncountable sets. The undecidability theorem of Gödel, which proves the existence of true propositions whose truth cannot be proved, occurs only in the logic of the second order. Shortly, the first-order logic applies to the study of simple and well-ordered worlds, of describable sets, such as the things that we can directly observe, while that of second-order logic applies to processes and behaviors that can evolve in an unpredictable way and indescribable sets of possibilities. This fact is directly related to the halting problem of Turing.

IV. The dual nature of mathematics.

From a purely formal point of view, any algorithm has a structure that can be encoded in an axiomatic form, so that the general properties of the algorithm can be logically deduced by means of proofs. In saying this, it is not excluded that non-demonstrable theorems exist. For example, the geometric algorithm, which makes physical use of ruler, pencil and compass, has the structure described in axiomatic form by Euclidean geometry intended as a logical system aimed at demonstrating geometric truths (for example the Pythagorean theorem).

The axiomatic theory of arithmetic was formulated by Peano and Dedekind in the second half of the 19th century. It is based on the primitive notions of *number* and *successor* and on the enunciation, in terms of these, of the axioms and rules of inference that establish the possibility of executing as many as you want additions and multiplications, even infinite operations and also the right to enunciate a general truth by means of an infinite concatenation of particular truths (*induction axiom*). The logical formalization of the arithmetic algorithm, extended to real numbers, is nothing but elementary algebra (the induction principle applied to geometry is implicit in Archimedes' method of exhaustion).

In summary, an algorithm can be described in two different ways: 1) as a repertoire of operations that can be concretely carried out over time on the objects of a certain sort; 2) as a system of axioms that describe the rules of computation and the properties of the classes of objects on which the algorithm can operate. To fix the ideas, let us limit ourselves to the arithmetic.

In the first way it can be described by a language whose universe of speech has *particular integers* and the operations to be performed on them. We can call it the universe of *arithmetic language*. In the second way it is described by a language that has the language of the arithmetic algorithm as the universe of speech, and therefore we can define the metalanguage of the arithmetic language. The objects of this *metalanguage* are the *general properties of integers*, or, in other words, the *classes of integers*, since the characterization of a property is equivalent to the definition of a class, and the possible operations on them. Similarly, every other algorithm presents itself, on the one hand, as a system of procedures designed to concretely transform a set of particular data into another set of particular data; for the other as an abstract formal structure that defines and elaborates an infinite collection of general properties.

The distinction between the *operational* and the *logical* aspect characterizes the whole of mathematics in a dualistic way. Historically, a certain rivalry has been established between the two aspects, which manifested itself, first, in the attempt, mainly pursued by Gottlob Frege (1879-1903), to reduce arithmetic to logic, then in Gödel's investigation aimed at research of the logical difficulties encountered in describing the properties of the arithmetic algorithm. The comparison of the two points of view had the effect of expanding the logic beyond the limits of their original formulations (through the works of Whitehead, Russell, Zermelo, Fraenkel, Hilbert, Tarsky etc.) and then the theory of algorithms (with contributions from Church, Turing, Kleene, Moore, etc.).

V. The self-referential circuit of Gödel.

Putting together all the things debated in the previous chapters, we can assert that the reasoning of Gödel is based on the following facts: 1) *the arithmetic is a universal algorithm, i.e. an algorithm capable of interpreting and translating any other algorithm*; 2) *the logical proofs of an axiomatic theory can be encoded into integers and translated into a set of arithmetic formulas*. The key step consists in this: *if the arithmetic encoding of a logical proof is applied to the axiomatic theory of the arithmetic itself, we obtain a self-referential circuit rich of extraordinary implications and consequences*.

This is explained by considering that an axiomatic theory is a way of producing a set of data starting from a set of data, which is just what the arithmetic calculus does, and that the operations of the Boolean logic, of which a proof ultimately consists, are similar to algorithmic elaborations of well-formed logical propositions. Since arithmetic is a universal algorithm, it is capable of enumerating in an orderly way all the axioms and the well-formed propositions of any mathematical theory. Since the logical proofs are similar to algorithms, the arithmetic is also capable of representing how the theorems of any logical theory can be derived from its axioms (for example, those of Euclidean geometry or rational mechanics), or from other theorems of the same theory, in the form of arithmetic operations carried out on the numbers which encode axioms and theorems.

The dance of complexifications and simplifications that intervene in a logical proof will be correspondingly translated into a sequence of elongations and shortenings of numerical expressions. Proceeding in this way, we will be able to represent the entire system of logical reasoning admissible by the theory so encoded as a system of arithmetic operations.

Therefore, the arithmetic algorithm is put in the position of interpreting itself and proving, in the form of particular arithmetic calculations, all its logically provable properties. So, for instance, the theorem "there are infinite primes" – a proposition that cannot be verified by analyzing case by case which numbers are primes – will be encoded in a particular number, which may be very large. The proof of the theorem will then be introduced in the form of a succession of arithmetic calculations which relate this number to those that encode the axioms. In this way, arithmetic – not as an *algorithm* but as an *axiomatic theory* – is put in the position of encoding all its *general* properties in form of *particular* arithmetic calculations.

This self-referential circuit can be extended to all that is provable in the framework of arithmetic. More generally, we can assert that, through this *gödelization* procedure, the arithmetic language manages to incorporate in its universe of the discourse all the *logically provable* propositions of its metalanguage.

VI. The explosion of indescribable complexity.

By definition, two sets that can be mapped on each other have the same *cardinality*. In the case of finite sets, the cardinality is equivalent to the number of elements. However, the notion of cardinality also extends to infinite sets. An infinite discrete set can be mapped into one of its parts in infinite different ways. For example, all integers can be mapped to even numbers by multiplying all of them by 2. Therefore, a set is infinite if and only if it has this property. It can be shown, for example, that all the points of a square can be mapped onto one of its sides. However, in many other cases maps of between infinite sets cannot be carried out. Thus, for example, as Cantor showed in 1867 by applying the diagonal method, while integers can be mapped into rationals, and vice versa, a one-to-one correspondence cannot be established between the points of a segment and the set of natural numbers. This means that the cardinality of the continuum is higher than that of the discrete.

Let us transfer these concepts to the Gödelian analysis of arithmetic. Since the totality of possible arithmetic calculations can be ordered and enumerated, the language that has as its object all possible arithmetic expressions, has the cardinality of the discrete. However, the language that has as its object the propositions that describe the possible classes of arithmetic calculations, i.e., the arithmetic metalanguage, has the cardinality of the continuum. Being this the state of affairs which case, we can easily realize that there must be infinite propositions of the arithmetic metalanguage that cannot be encoded in the arithmetic language.

We reach the core of Gödel's investigation when we realize that the cardinality of the arithmetic metalanguage is higher than that of the arithmetic language. Since the construction of Gödel maps the logically demonstrable part of the arithmetic metalanguage in the arithmetic language, we deduce that the arithmetic metalanguage has infinite *undecidable propositions*, i.e. unprovable theorems, (if they were unprovable their proofs would be indescribably complicated). Gödel himself provided a concrete example of an undecidable proposition.

VII. Equivalence of Gödel's and Turing's views.

In origin, Gödel's analysis was concerned with the relation between the logical structure and the algorithmic power of arithmetic. The problem of this relation reflected the interests matured in the scientific debate during the second half of the nineteenth century and the first years of twentieth century. Gödel's results were understood and reinterpreted by Turing in 1936 as a problem regarding the relation between the structure and the behavior of a computing machine. The theory of Turing machines reduces the substance of this problem to its simplest form.

Despite its structural simplicity, the universal Turing machine can do everything that can be done by any other computing machine. This property ultimately depends on the fact that the structure of any more complex machine is encoded in a particular set of data of the Turing machine, i.e., its computation program.

Tanks to this reduction in structural complexity, the whole theory of the computing machines reduces to the problem of how a universal Turing machine with low structural complexity can work. In practice, the universality of the Turing machine identifies with its capability of carrying out any sort of computation, precisely as the universality of the arithmetic algorithm identifies in the computational capabilities of the universal Turing machine.

The analogue of Gödel's undecidability theorem is that a universal Turing machine is always capable of simulating any other machine; but that, in general, it is not capable of predicting whether a calculation carried out by another machine will stop after a finite number of steps (*halting problem*).

VIII. Constructive universality and automata capable of self-reproducing.

As already explained in section 1.2, von Neumann's theory of automata retraces the same conceptual scheme. In some respects, it represents a natural extension of Turing's theory.

A *constructing automaton* is a machine capable of using elementary objects available in its environment to produce more complicated objects to be stored in the same environment. To accomplish this, it must contain a detailed list of instructions and possess repertoire of sensors and working tools a rich enough to perform all the necessary sorting and assembly operations based on such instructions; thus, in particular, it must also be capable of processing information.

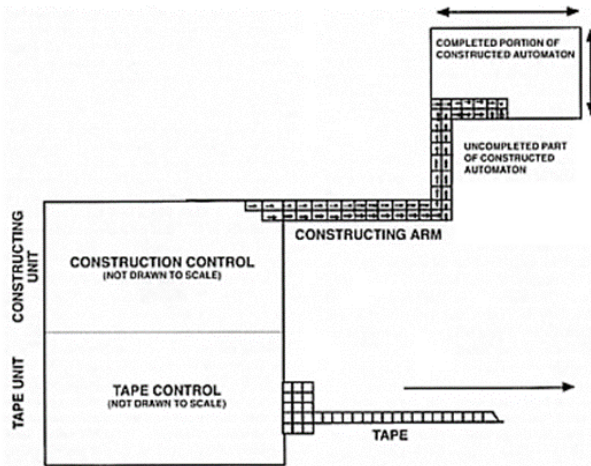


Fig.8. Simplified schema of the Universal Constructor described by von Neumann in *The Theory of Self-Reproducing Automata*.

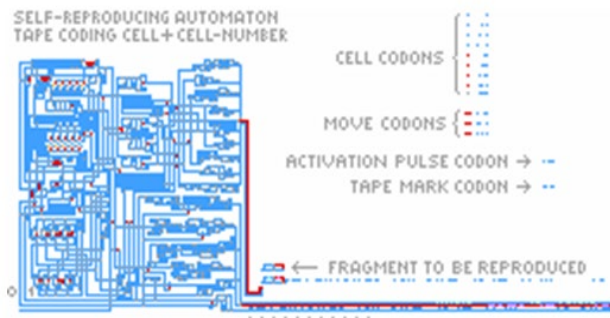


Fig.9. 32-state planar universal constructor capable of self-reproducing indefinitely (improved version of the automaton described in Fig.8.). The program of this automaton was implemented by the author of this paper in 2005.

In general, an automaton with small structural complexity will be only capable of constructing objects of complexity lower than its own; but if it is equipped with organs capable of performing a sufficient number of elementary operations, and its interior is sufficiently well organized, then it becomes a *universal machine*. That is, a machine which, by reading a list of instructions, becomes capable of constructing objects of any sort, even an automaton of complexity equal or greater than its own; therefore, also a copy of itself. It is then clear that a living cell has all the requirements of a universal automaton!

Von Neumann, just referring to the undecidability theorem of Gödel, argued that an automaton cannot “self-copy” by directly examining its own internal structure by means of sensors or other hypothetical introspection apparatuses. In fact, no measuring apparatus can perform a measurement within itself. In order for self-reproduction to take place, it is essential that the constructing automaton possesses a description of his internal structure and of a decoding apparatus that allows it to produce a detailed list of procedures or, equivalently, of a complete list of instructions concerning

the methods of construction and assembly of its parts. Such is precisely the genetic code of a living organisms. We can recognize here the analogue of the self-referential circuit of Gödel described in point V of section 1.2., without which the arithmetic algorithm could not be put in the position of encoding all its general properties in form of particular arithmetic calculations.

It is noteworthy that von Neumann arrived to assert all this about three years before that Watson and Crick discovered the DNA (1953), thus implicitly suggesting (for reasons related to a principle of optimal functionality) that the genetic information is encoded in linear data sequences (such as the symbols on the tapes of Turing machines). The great mathematician provided the example of a planar self-reproducing automaton made up of equal square cells, each equipped with 29 internal states, which can jump from one state to another according to certain transition rules and in ways dependent on the states of the former nearby. The crucial point of this deep vision, which can be regarded as the founding instance of theoretical biology, is the role played by the distinction between *structural complexity* and *behavioral complexity* of a system with a finite number of states.

Below a certain level of structural complexity, the system (the algorithm, the computer or the automaton) cannot evolve in a very complicated way. In these circumstances, the calculation capability of the algorithm, the behaviors of the computer or those of a constructing automaton appear predictable and describable in a finite amount of time. This means that the interpretative power of the algorithm, the simulation capability of the computer or the construction capability of the automaton are limited. In these circumstances, algorithmic, computational constructive universality are impossible.

But if the system has a sufficiently ordered and articulated structure, and is constructed in such a way to allow sufficiently complicated sorts of recursion, a novel and impressing fact occurs: the interpretation power of the algorithm, the simulation capacity of the computer or the construction capability of the automaton become universal. In these circumstances, the variety of qualitatively different behaviors of the system explodes exponentially reaching the cardinality of the continuum, and then it happens that the overwhelming majority of these behaviors are not predictable and describable on the basis of the knowledge of the internal structure of the system. The only way to know them is to observe directly their objective development.

IX. Languages, dialects, interpretative universality and creative power.

In conformity with the notions introduced so far, we can say that a language is universal if it can translate well the texts of any other language, despite the differences in grammar, syntax and semantic fields. The great national languages – such as English, French, Italian, German etc. – are *universal languages*. Instead the dialects, due to their grammatical, syntactical and semantic poverty, cannot translate sufficiently well the texts of a universal language (for example, despite my confidence with the Venetian dialect, I absolutely cannot imagine how the *Phenomenology of the Spirit* of Hegel could be translated into this dialect, so as preserve the original meanings and without producing irresistible comedy effects).

What marks the difference between a national language and the dialect is that the first, just because it is a universal language, is capable of perfectly describing its own grammar; that is, of being self-referential and therefore of functioning well and coherently both as a language and as a metalanguage.

The national languages were not born spontaneously from the practice of popular communication, but were created and refined through the patient and careful work of thinkers, writers, philosophers and scientists, who had the necessity of finding the most appropriate expressions for describing their complicated visions and culturally evolved thoughts. Now, as a universal algorithm is not only capable of translating any other algorithm, but also of producing data structures of unlimited complexity so, a universal language is capable of generating in a grammatically corrected way any sort of text, and expressing literary, philosophical and scientific concepts of unlimited complexity, greatly favoring the creative capacities of the thought. This explains the unlimited expressive power of the national languages compared to that of jargons and dialects.

However, this does mean that we must refuse the dialects. As the great Sicilian dramatist Luigi Pirandello (1867-1936) used to say: *of any given thing, the language expresses the concept, of that same thing the dialect expresses the sentiment*.

We can advance a similar argument also for artistic creativity. The substantial difference between an amateur painter and a truly creative one is that the former limits itself to copying or imitating a style, while the latter is provided with universal expressive and interpretative abilities that allow it to freely express all the sentiments, emotions and ideas arising in its mind.

Both in figurative arts and in music, the expression means and the technique of an artist form a language that, in the authentic artist, is untranslatable both in written language or in other form. The grammar and syntax of this language are personal creations, whose deep structure emerges only from the stylistic coherence of the works. The recursive character of the artistic work is essential for creativity. The artist tries and tries again, partly randomly and partly following some inspiration, selecting the results according to its aesthetic taste, until it obtains something that is perceived as self-consistent and provided with universal communication power.

1.3. The problem of self-consciousness.

An important example of self-referential Gödelian circuit is that which concerns the problem of self-consciousness. Several authors have observed that self-consciousness cannot emerge from a simple process of *introspection*. This is perfectly understandable since the thesis of introspection would give rise to infinite regressions of the kind: *I am aware of looking at the table*; $\rightarrow$ *I am aware of being aware of looking at the table*; $\rightarrow$ *I am aware of being aware of being aware of looking at a table*; etc. This situation is well represented by the image, shown in Hofstadter's book *I Am a Strange Loop* (2007), which represents the character of a comic of Ernie Bushmiller (1905-1982), named Sluggo, who reflects on himself (Fig. 10).



Fig.10. Sluggo.

This situation is in several aspects similar to that of the universal constructor of von Neumann that cannot copy itself by means of a self-inspection device, but can only do so with the aid of a computation program that encodes the inner structure of the constructing automaton. This similarity suggests that the “machine of the mind” also possesses a program that encodes its own *structure*. In this way, the entity that in the example given above shows up as “*I looking at the table*” would actually be the image created by a computer program that generates the image of a character, called *I*, who looks at the image of a table.

Incidentally, this sensation fits quite well with the witnesses of people suffering from a state of mental dissociation, who assert that sometimes they have the impression of observing themselves from the outside.

Telling this, we do not want to assert that every time we say *I* we refer to ourselves in flesh and bones, or to the objective neuronal activity of our brain, or to the subjective activity that emerges in

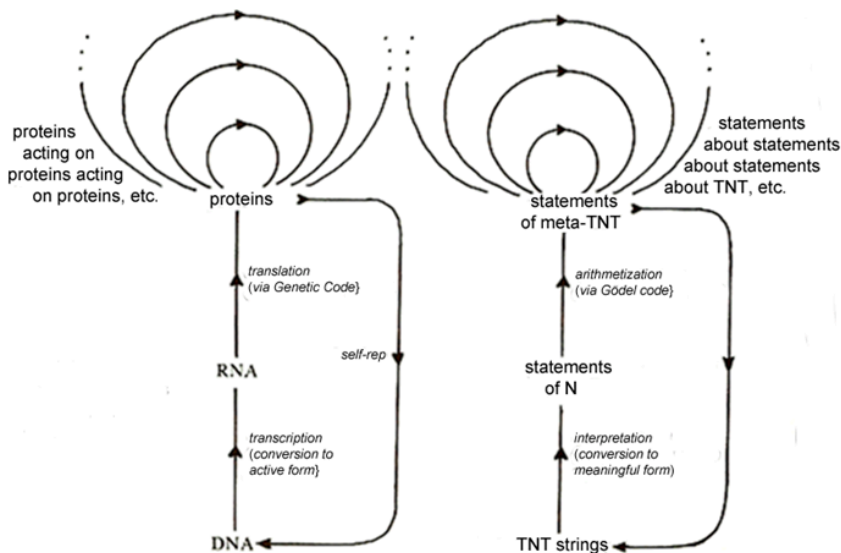


Fig.11. How Douglas Hofstadter sees the relationship between biological self-reproduction and logical-arithmetic self-reference.

our mind, but rather to a *representation of ourselves put in relationship with the representation of others and other things*. In homage to Hofstadter, who was the first to talk about this in his book *Gödel, Escher and Bach: An Eternal Golden Braid*, we call it the representation the *self-symbol*. The contribution given by Hofstadter is worthy of acknowledgement because this author was the first to introduce the idea that the mental process that generates the self-consciousness is very similar to the procedure used by Gödel in his research on

the self-referential capabilities of arithmetic. Basically, this procedure retraces the schema of the correspondence, indicated by Hofstadter himself (Fig. 11), between the self-reproducing capability of living beings and the reasoning through which Gödel arrived to discover the self-referential circuit of arithmetic.

In the vision of Hofstadter, the symbol of the self plays the role of the axiomatic theory of arithmetic that encodes the formal structure of the arithmetic algorithm, as explained in point V. In a similar way, the DNA of a living being encodes the biological structure of the cell that contains it.

In *I Am a Strange Loop*, Hofstadter specifies that the symbol of the self is generated by a recursive process instructed by a program that resides in the brain. In the same book, Hofstadter also arrives to the point of asserting that the process that generates the symbol of the self can also generate the symbols of the persons with whom the thinking subject has interacted and communicated for a sufficiently long time in the course of its life.

Unraveling the topic from metaphors, and a literary style reminiscent of *Alice's Adventures in Wonderland*, we can briefly describe his conception of the symbol of the self as follows: a thinking brain is the seat of processes that produce certain states of excitement of the nervous network, which the author defines *active symbols*. These are so defined because, unlike the passive symbols that belong to mathematical formalisms, they do not receive their meanings from relationships imposed by an external subject, but are themselves capable of "activating" and recruiting relationships with external objects. In order to maintain the analogy with the previous cases, we must assume that the "machine of the mind is capable of transforming the active symbols in other active symbols, according to procedures that are instructed in some unknown way.

Of course, no external supervisor must exist, no hidden conscious *homunculus* that reads and interprets a list of instructions for the purpose of making the brain to be able of performing these operations (Fig.12), otherwise it would occur a regression to infinity as that illustrated in Fig.12.

The activity of symbolic production and reproduction of what is used to call *mind* must generate and sustain itself autonomously through the causal interaction of active symbols, which are activated and deactivated in the brain according to the rules of a specific dynamic.

In analogy with the concepts illustrated in the previous paragraphs, we add the following definition: mind *A* is capable of *interpreting* mind *B* if *A* is capable of establishing a correspondence between each of its symbolic productions and a symbolic production of *B*.

In order for this to happen, *A* must possess all the means that are necessary to recognize and decode the symbols produced by *B*. It must therefore have the capacity of internal representation coordinated with devices interacting with the external world. A mind with insufficient structural complexity, like that of a dog or a monkey, cannot interpret anything other than symbolic productions of limited complexity. But if its structure is sufficiently complicated and articulated, and its memory is large enough, it becomes a *universal mind*. Provided that it be properly trained, it will be able to interpret any other kind of symbolic production: that of a computer, of an animal, and those of other human minds. It is then evident that for this peculiar reason *the human brain can be regarded a universal machine!*

Hofstadter does not explain how a human brain can generate a universal mind, since in the book quoted above, and in others of the same author, the universality of the human mind is implicit in the analogy of the self-conscious mind with the self-referential circuit of Gödel discussed in point V.

However, we can reasonably hypothesize that this self-reference capability is established in the earliest stages of the developmental age of the child, through the communication of body gestures, facial expression of emotional states, the exercises of their simulation, the episodes of emulative learning and the assimilation of the behavior schemes of the adults. All these developmental stages correspond perfectly to what Antonio Damasio (1945-...) describes in his books on the neurological basis of emotions. But, of course, the development of all these skills does not suffice to generate the capacity for universal interpretation.

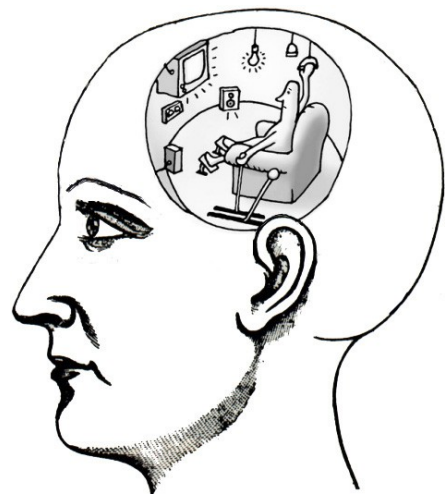


Fig.12. Homunculus nested in the brain.

Unfortunately, regarding the neurological mechanisms of self-consciousness, Damasio does not provide explanations sufficient to clarify the question of the capacity for self-reference of the mind. We can try to complete his vision by bringing into play the analogy with the self-referential circuit of Gödel construction and the theory of universal automata of von Neumann.

This topic is about the relation between two fundamental abilities of the human mind: the *imagination*, to be identified as the capacity to produce mental images and symbolic representations, as described above, and the *language*, to be identified as the means of expressing our mental states and the methods of interpersonal communication. It seems reasonable, in fact, that it is possible to establish a similarity between these two abilities of the human mind and the two ways of considering the arithmetic discussed in point *IV* of section 1.2: the algorithmic and the logical way. The Gödelization of the relation between imagination and language may be carried out as follows.

From an objective neurological standpoint, the activity of thinking is due to the interaction of a system of cortical areas and brain nuclei that is capable of recruiting and evoking the contents of the memory in the form of states of collective excitation of neurons. On the other hand, from the subjective point of view of the mind, it appears as a complicated elaboration of particular fantasies and specific behavioral decisions of the thinking ego. Regardless of the ways by which the contents of memory may have been originated from sensory perception, the elaboration of a fantasy has the formal properties of an algorithmic recursive production. Since in this case it is a recursive production of mental states, it can reach levels of indescribable complexity that cannot be summarized in any way by a particular mental state, but only interpreted and understood by another mind capable of carrying out an analogous recursive process, hence a common language.

Of course, this is possible only if each mind has an apparatus for symbolic production and a communication system similar to those of the other. Hence a common language. No mind can directly perceive the fantasies of another mind, but it can reconstruct them in its own imagination provided that the messages received by the other mind are sufficiently coherent and detailed.

A mind capable of understanding the fantasies of any other mind and communicate them to any other mind, can be called a *universal mind*.

Now, as a universal algorithm is capable of self-referencing, so a universal mind can interpret and understand itself. In order for this can happen, imagination and communication must be two independent activities which enter in mutual relationship through the interpretative processes of thought. Taken together, they impart a dualistic character to mental activity similar to the one between arithmetic algorithm and arithmetic theory. Self-consciousness, in its essence, consists precisely in the gödelization of thinking activity. The self-reflexive process of the mind is capable of producing the linguistic messages which describe the free fantasies of the mind; conversely, the fantasies produced by the mind are in turn capable of evoking linguistic messages.

But in order for the Gödelian circuit of self-consciousness to close, the communication system of the mind and the production system of mental images must be mutually compatible; in other words, the language must be coherent with its metalanguage; what is said must describe well what is imagined.

We should therefore hypothesize that in a universal mind there exists a special symbolic system capable of representing and exemplifying imaginatively even the formal structure of its own thinking activity, the relationships between the parts of this structure and the functions of its internal mechanisms. By generalizing the notion of "self-symbol" introduced by Hofstadter, we can call it the *symbolic system of the self*.

In the Turing machines and in the automata of von Neumann, interpretative universality depends on the possibility of reproducing and simulating in their interior the processes of other machines. In the same way, the universality of a mind depends on the possibility of representing the behavioral stereotypes of other minds, as in a sort of puppet theater.

As a universal Turing machine is capable of simulating any computer and a universal von Neumann automaton is capable of constructing all that any machine can construct, so a universal mind is capable of imagining and describing all that any being provided with a brain can imagine and communicate.

In a sense, this hypothesis does nothing else but reintroducing in another form the homunculus nested in the brain. Only now this strange character must not be imagined as a neuronal structure located in a special part of our brain nor be identified with the somato-sensory and motor homunculi, which are used to paint on the maps of the cerebral cortex to indicate where neurons reside which correspond to the different parts of the body (Fig. 13).

The somato-sensory and motor structures are an aggregate of specialized areas constantly engaged in coding and conveying mutually independent information flows between brain and soma. The homunculus that is needed must be understood as a complex symbolic entity that represents in an integrated way the pattern of what happens in the brain, not the actual mental processes in their complexity. In the same way, the axioms of theoretical arithmetic do not represent all that the

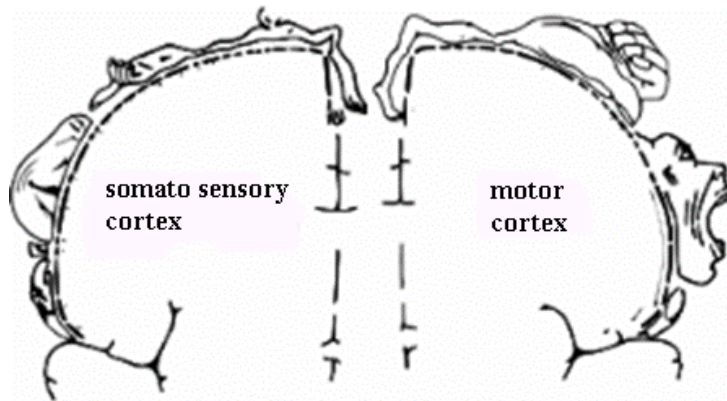


Fig.13. Somato-sensory and motor homunculi nested in the brain

arithmetic algorithm can generate, and the double helix of DNA does not represent what happens in the living organism.

The great dream of AI, is the creation of artificial self-consciousness, therefore seems to be strongly conditioned by the possibility of constructing computers capable of communicating linguistically with human beings, interpreting their thought, introjecting their structural scheme, up to point of developing their own symbolic system of the self and thus

becoming capable of self-interpretation. Can modern computer technology go that far?

1.4. The introspective machine of Konolige

The self-referential capacity of arithmetic suggests that a universal Turing machine can work in a self-referential way, that is, be put in a position to analyze, describe and communicate the calculations that it has carried out or is carrying out and the procedures he has used or is using to obtain its results. A computer capable of doing this could be called *introspective*, although this denomination is considered improper by some authors. It would provide a model of the self-reflexion capacity of human thought and its implementation would amount to the creation of artificial thought. In the 1980s various researchers undertook some research aimed at developing a theory of introspective machines (Bartlett, 1992).

In an article of 1985, Kurt Konolige defines the *introspective machine* as a system provided with a *subsystem of beliefs*. According to its definition, a subsystem of beliefs is a computational structure within an artificial agent capable of representing a finite number of beliefs concerning the facts of the world. This subsystem is capable of accepting a question asked by an external agent, and attempting to verify whether the answer to the question can be derived from its system of beliefs. The operating principle of the machine is based on the decomposition of a complex question into simpler questions and on the subsequent attempts to answer these sub-questions.

Konolige assumes that the introspective machine possesses for each of its beliefs C , which he defines as *non-doxastic*, also the belief "*I believe that C is one of my believes*", which he defines as *doxastic*. This means that doxastic language is the metalanguage of non-doxastic language. Here the analogy with the condition that the Gödelian circuit of self-consciousness be closed, discussed in section 1.3, is evident. Clearly, a necessary condition for the machine to be really capable of introspection it is necessary that the algorithm that processes non-doxastic questions be *universal*. Gödel's undecidability theorem then ensures that there are infinite questions to which the machine cannot answer.

Konolige does not say it explicitly, but another condition for the machine to really possess the introspective capacity is that it is capable of autonomously asking itself all the questions that can be asked by an external agent. Can a Turing machine be programmed to do all this? The answer is negative. The capability of generating autonomously questions that can be unpredictably posed by external agents contrasts with the determinism of the Turing machine.

This capability could be supplied by a device capable of asking "well-formed" random questions on the basis of an appropriate generative grammar. The difficulty lies in the fact that no deterministic machine can really produce random sequences of symbols.

The question must therefore be rephrased as follows: can a universal Turing machine, possibly equipped with a source of random questions, be programmed to function as an introspective machine? *In theory yes*, because the universal Turing machine satisfies all the conditions for its gödelization. *In practice, no*, because the computation times of the recursive procedures, by means of which it can try to answer questions – by building and analyzing classes of beliefs – generally increase exponentially with the number of beliefs.

This depends on the fact that the cardinality of the classes of beliefs is higher than that of the beliefs. The bottleneck that prevents the introspective process in practice lies in the fact that the Turing machines, and with them every ordinary type of computers, are *sequential*, in the sense that they are built so as to perform a relatively small number of operations at each cycle ruled by their internal clock. The awareness of a mental event would serve us very little, and perhaps it would be an obstacle, if instead of arriving with a delay of about half a second after the brain unwittingly elaborated the decision – as the experiments of Libet and collaborators (1979) have shown – it took longer to produce its effects on the behavior of the thinking subject and on communication with other thinking subjects. Its accomplishment requires that the machine of the brain is not only very powerful but also very well organized to generate introspective processes in short times. This is perhaps the cause of its late appearance during the evolution of the species.

1.5. The computation power of the brain

The main drawbacks of ordinary computer and of the Turing machine, which represents them all in a paradigmatic way, are the limited number of operations that can be performed in one millisecond (about the duration of a nervous pulse) and the seriality of the computations. This means: *a)* that the time taken by these machines to perform a set of independent calculations cannot be less than the sum of the times necessary to perform them individually; *b)* that only processes whose computation times grow as a small power in the number of initial data are possible in practice.

At the end of the last century a desktop computer with a 32-bit microprocessor could perform about four billion operations per millisecond during one microprocessor cycle, but today one with 64-bit microprocessor can perform thousand times more, i.e., about 4×10^{12} operations per millisecond. The speed of the computers depends on the fact that electrical signals propagate at speeds close to those of light, while the speed of nerve signals, being limited by the electrochemical nature of the nervous processes, has the same order of magnitude of the sound speed. This slowness – in the face of the evolutionary advantage of mammals – has forced the nervous systems of these animals to adapt to a principle of *maximum parallelization possible for the same functions*. In the struggle for survival, the parallel information processes carried out in their nervous systems have become much more efficient than those that could be carried out – in a more precise but slower way – by serial computing machines. Until a few decades ago, many neurophysiologists seemed to agree that the effective processing units of nerve information in the cerebral cortex are not individual neurons, but small populations of excitatory and inhibitory neurons comprising a few thousand neurons. The electroencephalographic (EEG) activity, which is mainly recorded in the frequency range 6-100 hertz, was interpreted as an oscillatory state of the firing activities of these neurons due to the delayed feedback of the inhibitory neurons on the excitatory ones.

Knowing what exactly EEG activity is relevant because this would allow us to understand what are the mechanisms of neuronal dynamics and their function in cognitive processes.

One of the most widespread theories is that of *dynamic chaos* supported by Walter Freeman (1999), who interprets the oscillatory activity of the brain as the manifestation of a dynamic in which periods of disordered activity alternate with processes of spontaneous organization towards precise space-time configurations (*attractors*) rich in information.

Unfortunately, this theory has many flaws. First of all, it is known from computational simulations that systems of weakly interacting oscillator can reach a state of synchronization only in a time not less than that of several oscillation cycles.

Thus, for example, a system of neuronal oscillators operating at the frequency of 80 hertz (*gamma activity*) would take no less than half a second to produce a configuration of neurologically significant activity.

While neurons in the visual areas are known to synchronize in about one millisecond (Singer, 1999). Furthermore, identifying the functional units of the brain with small populations of neurons would lower the effective structural complexity of the brain by a few thousand times. Since the human brain contains about ten billion neurons (10^{10}), we infer that the brain should house around ten million such small units. Even working at the average oscillation frequency of 40 hertz, the brain could perform at most 400 million operations per second.

Therefore, we would deduce that the computing power of the human brain is approximately equal to that of a modern desktop cal.

But in the comparison between the performance of the brain and that of the computer, the most serious differences are those concerning the memory capacity and the relationship between the access time to memory and the time of the computation cycles. These differences are vastly more unfavorable for the calculator.

The memory capacity of a neuronal synapse can be estimated at around 1 byte (8 bits). Given that each neuron has an average of ten thousand synapses, we can estimate that the memory capacity of a neuron is about 10 kilobytes and that of the thalamus-cortical circuit and some tens of terabytes (thousands of billions of bytes). Probably much more if other brain cells (in particular astrocytes) also have memory mechanisms.

Regarding access speed to memory, that of a synapse can be estimated 1-2 milliseconds (equal to the activation time of a synapse), which is a negligible fraction of the average time of one cycle of a neuronal reverberating circuit.

As for the external memory of desktop computers, it must be considered that today hard disks are on the market that have a memory capacity of a few terabytes (one thousand billion bytes), an amount comparable with that of the brain.

Unfortunately, the access time to data stored on a hard disk is in the order of one millisecond. This greatly delays the effective calculation speed of the microprocessors. In fact, to perform a calculation operation using data stored on the hard disk, you must first read the data from the disk and then write the results on the disk (*swapping*). I think everyone who has a computer with little random-access memory (RAM) knows how tedious these swapping operations are.

By contrast, the RAM memory, which is a totally electronic, has an access time of about two machine cycles, which is therefore not negligible compared to the computation cycle of the microprocessor. This in practice decreases the effective time of the computation cycle by five times ($2+2+1$).

In view of the future advances in technology, we can presume that if the RAMs were as much as that of a hard disk, the effective computing power of a computer would be very close to that of a human brain. Unfortunately, however, the RAMs now available are at the most of the order of magnitude of a few gigabytes (billions of bytes).

These figures give us the measure of how far is the possibility of simulating the functions of the human brain is with a desktop computer. They also explain why the visual system is able to recognize an object placed in a complex environment in a tenth of a second, while a computer, to perform this function with an acceptable degree of efficiency, would take thousands of times longer.

However, we must consider that technological progress seems to know no limits, so it is not risky to think that in the not too far future it will be possible there will be information processing systems capable of simulating the parallel processes of the human brain in real time.

It is therefore legitimate pose the question of whether it will be possible to produce a "self-conscious" automaton.

Therefore, the substantial difference between the brain and the computer lies in the enormous difference in the relationship between the amount of data that can be read and written in a cycle of EEG activity and in a cycle of the microprocessor CPU: a few billion bytes in the first case, a few tens of bytes in the second case. This huge difference is due to the fact that the information processing of a computer is serial, whereas that of the brain is parallel.

1.6. Serial and parallel processes

Is there really a big difference in efficiency between parallel and serial machines? And if there is, how can this be explained?

In recent years, electronics and information technology did an attempt to confront in a systematic way, both in theory and in practice, the problem of building parallel computers. The difficulties encountered have shown how far computer science is still far from being able to simulate the machine of the mind.

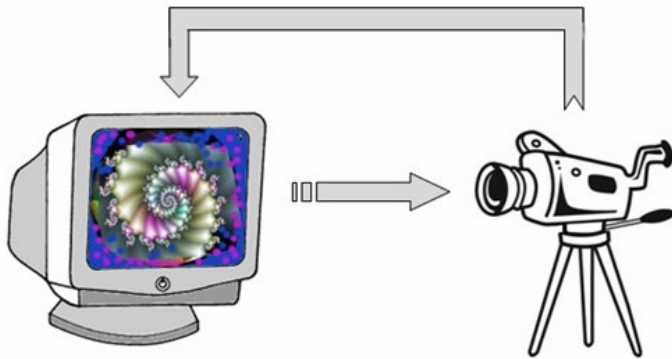


Fig.14. Recursive video-television process. The circuit generates complex abstract images that evolve unpredictably.

Serial machines transmit and receive data sequences through a few communication lines. However, when they operate recursively, they are capable of producing results of indescribable complexity. The brain, on the other hand, works with very dense data flows that are transmitted and received simultaneously by spatially extended apparatuses: the visual, tactile, muscular sensory apparatuses, the areas of the cortex, the basal ganglia, the brain nuclei, etc. and all the organs that make up the soma and appear to lack sufficient precision to function recursively.

Can a serial machine perform a parallel processing process equivalent to that of a parallel machine relying on its higher speed? Conversely, can a parallel perform recursive processes of complexity equivalent to that of a machine, counting on its greater complexity? Which of the two types of machines is able to provide the best performance for the implementation of the self-referential capacity of thought?

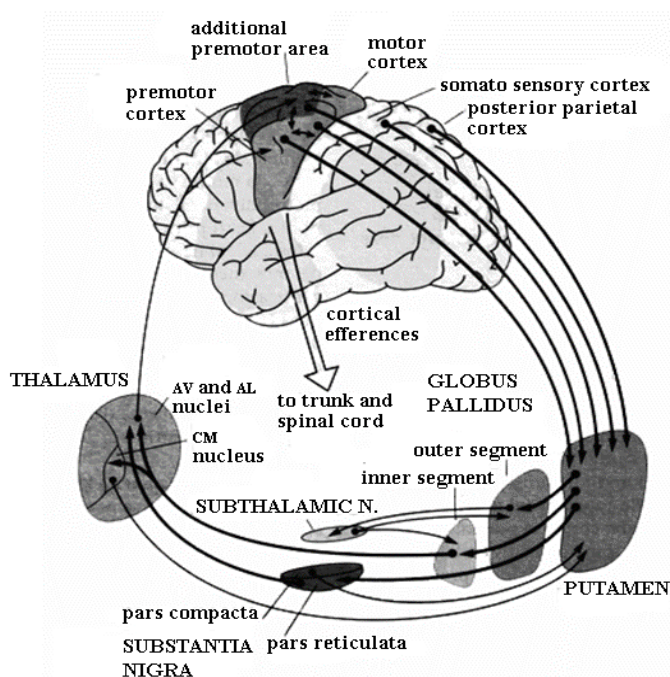


Fig.15. The thalamus-cortex circuit.

With a fast scanning procedure, any set of synchronous data can be easily converted into a sequence of bits. Conversely, any sequence of bit can quickly be transformed into a synchronous set of data. Cameras and TV screens do just that. Because of the speed of electronic scanning, the computers are capable of receiving and sending flows of parallel data in a way similar to that of the human brain.

In *I Am a Strange Loop* (2007), Hofstadter indicated the circuit shown in Fig.14 as an example of a parallel recursive process potentially capable of generating images of great complexity: a camera is set coaxially in front of the display of a television, which in turn distorts and captures and the image produced by the camera, arranged in such a way to preserve dimensions of the image.

Although the process is entirely carried out by serial devices, it provides an example of how recursive processes of the brain may work. For example, we can assume the thalamus-cortical circuit illustrated in Fig.15 is similar to the camera-display circuit described above, with the thalamus in the role of camera and the brain cortex in that of the display.

It is evident that the structural complexity of the brain is incomparably greater than that of the video-television circuit. Can more sophisticated serial processes fill the gap?

1.7. The problem of the serialization of parallel processes.

In order for a serial machine to compete with a parallel machine, it must be capable of carrying out a very long sequence of logical-arithmetic operations on the sets data coming to input.

This situation occurs, for example, in the coding and decoding procedures of video signals, which are currently applied upstream and downstream of the data transmission systems to increase the resolution of the images reproduced by the screens.

The analogic information flow produced by the camera is first converted into an array of digital signals and subjected in this form to a compression procedure (*zipping*), then transmitted in this format to the video receiver, and finally decompressed and sent to the display.

To give an idea of the problem we are trying to solve, suppose we use a good quality desktop computer to transmit a film that in an uncompressed format occupies about one megabyte (8 million bits) of memory, with a small delay per frame, at the rate of 24 frames per second. To compress this massive flow of information in the most efficient way, we must analyze the correlations between the parts of each frame and between a sequel of frames and re-encode the information so as to drastically reduce the elements of redundant information.

This process requires that large amounts of data be read and written very quickly in a temporary working memory. Since the minimum data access time of a hard disk is of the order of one millisecond, one cannot think of using hard disks to perform these reading-writing operations. Fortunately, electronic memory, or RAM (*random access memory*), is one hundred thousand times faster, although its capacity is thousands of times less than that of a hard disk. So, it would allow to perform the work of compression and decompression of the information contained in the film approximately in real time. However, with this type of working memory, the serial computer would be capable of simulating parallel recursive processes with information flows comparable, at most, to those of insect nerve systems.

In short, the substantial difference between a serial machine and a parallel one like the brain – if this were capable of operating recursively – is that the former must perform an enormous number of calculation cycles to obtain what the second could do in a few cycles.

A distinctive feature of serial computers is the clear separation of hardware from software, which means separation of data from programs in the memory systems. This separation responds to the requirement for optimization of serial processes. The way of executing a serial process depends both on the architecture of the microprocessor, which is fixed, and on the logical-arithmetic computation programs, which are part of the data stored in the memory and vary from case to case. The difference between programs and data lies only in how these types of data are used by the microprocessor.

The computer science of parallel processes identifies, in practice, the parallel computer with a set of serial computers that exchange messages (Codenotti and Leoncini, 1990). Our interest is turned instead to data processing systems in which the parallelization is maximum compatibly with the condition of maximum efficiency. It is plausible that the theoretical principles that stand at the basis of parallel calculation are different from those on which the theory of serial computation was based.

We will try to set the problem without pretending to arrive at a general solution. In the following we will mean by parallel computer a computer whose calculation processes are massively parallelized. Since calculators of this type do not exist for now, the best you can do is to try to understand what features they should have.

In order for a parallel machine to be able to compete with a serial in carrying out recursive processes, it must be able to maintain the synchronization status of the information flows with great precision, although these can be divided into variously specialized components that cross different and distant areas of the car. The recursive condition requires that parallel information flows pass through closed stages of stages while maintaining the synchronization state and that therefore the connections between the different parts of the system are reciprocated by connections that proceed in the opposite direction. It is a fact that almost all connections between the operative parts of the brain are reciprocated by reverse connections (Asanuma and Crick, 1986) and that there are sequences of closed loop stages in the brain that could be devoted to the recursive processing of information: the Papez circuit and the extrapyramidal circuit (Fig.12) are the two most evident (Nieuwenhuys *et al.*, 2008).

However, the construction of massively parallel computers encounters an apparently insurmountable difficulty that the animal brain seems to have efficiently solved: a very large number of computing units must concurrently contribute to the same overall process, whose efficiency depends strongly on system architecture.

Now, since the addressing of data is provided by internal connections, and that the execution of a program requires that the data be addressed in unpredictably various and different ways, it is easy to understand that the execution of a program requires continuous modifications of the system architecture, or the use, if necessary, of variously specialized modules, towards which the parallel information flows must be quickly directed, activated and deactivated, since in general an architecture that is optimal for a certain process may not be optimal for another.

In the first case, the question arises of which information flow control techniques and which methods of rapid reorganization of the architecture can be used to obtain high computation efficiency. In the second case, how vast the repertoire of specialized modules should be to satisfy the calculation needs. But in both cases, we can say that the programs tend to coincide with the architecture of the system.

Furthermore, if we request that the calculator be universal, the capability of quickly and incessantly reconfiguring its architecture becomes indispensable. A universal parallel calculator cannot have an optimal architecture only for a particular process. It must be versatile enough to perform a wide repertoire of processes arranged in unpredictable sequences, and therefore change quickly the architecture of all its parts in a coordinated way. How can all these functions be implemented in a parallel machine?

This problem has certainly been solved by animal brains, given that natural selection has always provided for the discovery of optimal solutions for all the functions of biological systems.

The human brain is certainly a universal parallel machine capable of performing recursive processes of indescribable complexity in a few moments. How else could one explain, for example, the complexity of dreams? So how does the brain manage to do what seems so difficult for an artificial machine? A question then arises: is it possible to build a computer capable of simulating the human brain?

The answer is: the simulation of the human brain is impossible because the complexity of this organ is indescribable, being itself the product of a recursive process, partly phylogenetic and partly embryogenetic, which lasted half a billion years.

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